Capstone Project - Walmart Sales Forecasting

Submission Report Draft

Prepared from forcast.ipynb and Walmart.csv

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1. Problem Statement

A retail organization operating multiple outlets needs to align inventory and operational planning with demand. Historical weekly sales vary across stores and may be influenced by holidays, temperature, fuel prices, consumer prices, and unemployment. The project addresses two questions:

1. What patterns and relationships in the historical data can support store-level business decisions?
2. Can weekly sales be forecast for each store for the next 12 weeks?

2. Project Objective

The objectives of this project are to:

- Assess and clean the Walmart weekly-sales data.
- Identify relationships between sales and economic, weather, and holiday variables.
- Compare store performance and identify the highest- and lowest-performing outlets.
- Build and validate a separate time-series forecast for each of the 45 stores.
- Evaluate WMAPE, RMSE, and MAE, then improve performance through feature selection and hyperparameter tuning.

3. Data Description

Walmart.csv contains 8 columns and 6,435 rows. Each row represents one store?s weekly observation. The dataset includes 45 stores and 143 weekly observations per store, covering 5 February 2010 to 26 October 2012.

Feature	Description
Store	Store identifier
Date	Week-ending sales date
Weekly_Sales	Sales for the store in that week
Holiday_Flag	1 for a holiday week, otherwise 0
Temperature	Recorded temperature
Fuel_Price	Regional fuel price
CPI	Consumer Price Index
Unemployment	Regional unemployment rate

4. Data Pre-processing Steps and Inspiration

The preprocessing and analysis approach was inspired by the need to create a reliable, chronological store-level forecasting dataset.

- Loaded the CSV with pandas and inspected structure, data types, and descriptive statistics.
- Converted Date to datetime using day-first parsing because the source contains different date representations.
- Checked null values, duplicates, column types, and summary statistics.
- Found no missing values and no duplicate rows in the supplied dataset.
- Verified that every observation occurs on Friday and every store has the same 143-week date range.
- Created a sales-in-millions measure for visualization and store comparisons.
- No additional feature engineering was required because observations are consistently weekly and aligned across stores.
- For forecasting, sorted each store chronologically and reserved its last 12 observations for final testing.

5. Choosing the Algorithm for the Project

The selected algorithm is Prophet, applied independently to each store's weekly sales series. The model uses the date as the time variable and Weekly_Sales as the target. The configured model uses yearly seasonality, disables weekly and daily seasonality, and generates a 12-week forecast horizon.

The final modeling workflow is chronological: earlier observations are used for training, while the final 12 weeks are held out for testing. This prevents future sales information from leaking into model training.

6. Motivation and Reasons for Choosing the Algorithm

- Prophet is designed for business time series containing trend and recurring seasonal patterns.
- It can model annual seasonality that is relevant to holiday-driven retail demand.
- It provides a consistent workflow for fitting many independent store series.
- It supports practical hyperparameter tuning for trend flexibility and seasonality strength.
- Its forecasts and components are relatively interpretable for business stakeholders.
- A store-specific model is appropriate because sales volumes and economic relationships differ considerably across stores.

7. Assumptions

- Historical relationships and seasonal behavior are informative for the next 12 weeks.
- Each store is modeled independently and has sufficient history for the selected approach.
- The weekly observation schedule and date alignment are valid.
- Future external regressors are not required for the final sales-history-based forecast.
- WMAPE, RMSE, and MAE are suitable for comparing stores with different sales volumes.
- Correlation findings are descriptive and do not establish causal relationships.

8. Model Evaluation and Techniques

The notebook follows a staged evaluation process:

3. Establish a baseline Prophet model for each store.
4. Test candidate feature configurations using time-series cross-validation with MAE, RMSE, and WMAPE.
5. Select the configuration with the best WMAPE behavior; the notebook reports that the baseline sales-only configuration performed well and was used for the final model.
6. Tune changepoint_prior_scale over 0.001, 0.01, 0.05, and 0.1; seasonality_prior_scale over 0.1, 1, 5, and 10; and additive versus multiplicative seasonality.
7. Refit the final model and evaluate forecasts against the held-out 12 weeks for all 45 stores.

Metric	Baseline average	Final overall result
WMAPE	6.55%	4.17%
RMSE	92,018.39	62,451.76
MAE	72,216.33	42,849.87

The final model improves on the baseline across all three reported measures. WMAPE is especially useful because it scales absolute error by total sales and is less dominated by the largest stores than an unscaled aggregate error.

9. Inferences from the Same

Exploratory findings

The overall relationship between unemployment and weekly sales is weak. Store 38 and Store 44 show the strongest negative store-level associations, approximately -0.785 and -0.780, making them the most sensitive to unemployment among the outlets examined.

The notebook identifies high-sales spikes around 26 November and 31 December. Average weekly sales were approximately $1.123 million in holiday-flagged weeks versus $1.041 million in non-holiday weeks.

The temperature-versus-sales regression visualization shows a negative slope. CPI effects vary by store: Store 36 has the strongest negative association, approximately -0.915, while Stores 38 and 44 have strong positive associations, approximately 0.813 and 0.740.

Store performance findings

Store 20 is the top performer at approximately $301.40 million cumulative historical sales, followed by Store 4 at $299.54 million and Store 14 at $289.00 million. Store 33 is the lowest performer at approximately $37.16 million. The gap between Stores 20 and 33 is approximately $264.24 million, or 711.08% relative to Store 33.

Business implications

- Demand is heterogeneous across stores; ranking and forecast error should be monitored individually.
- Holiday periods should inform inventory and staffing plans.
- Stores 38 and 44 warrant closer monitoring for unemployment-related sensitivity.
- The final 12-week forecast accuracy is strong for operational planning, but uncertainty and store-specific performance should be reviewed before committing inventory.

10. Future Possibilities of the Project

- Add known-future regressors such as promotions, markdowns, local events, store openings, and holiday calendars.
- Evaluate prediction intervals and store-level error distributions, not only overall averages.
- Compare Prophet with SARIMA, gradient boosting, tree-based regression, or ensemble forecasts using the same chronological validation.
- Retrain on a rolling schedule as new weekly sales arrive and monitor drift.
- Extend the dataset beyond October 2012 and include promotion, inventory, competitor, and store-metadata fields.
- Interpret yearly seasonal components cautiously for shorter histories; more data would improve stability.

11. Conclusion

This project delivers an end-to-end Walmart sales forecasting workflow. The data was validated, exploratory analysis identified store and holiday patterns, and a store-level Prophet approach produced 12-week forecasts. After feature-selection validation and parameter tuning, the final reported performance was 4.17% WMAPE, 62,451.76 RMSE, and 42,849.87 MAE across 45 stores. The forecasts can support demand planning when combined with operational information and refreshed as new observations arrive.

12. References

- Walmart.csv, supplied project dataset.
- forcast.ipynb, supplied forecasting notebook.
- Taylor, S. J., & Letham, B. (2018). Forecasting at scale. The American Statistician, 72(1), 37-45.
- Pandas documentation: loading, datetime conversion, grouping, and correlation analysis.
- Meta Prophet documentation: time-series forecasting and model configuration.
- Scikit-learn documentation: time-series cross-validation and evaluation metrics.